

DetectorPlane Causality in Quantum Measurement

A Benchmark Framework for Measurement Integrity and Control

This handout summarizes the framework proposed in **DetectorPlane Causality in Quantum Measurement**. The work treats quantum measurement as an instrumentation and signalprocessing problem and introduces the **Quantum Measurement Stack (QMS)** along with the **QMCTB01 benchmark** for evaluating detectorplane coherence preservation.

Causal Measurement Chain

Field Excitation → Imprint Formation → DetectorPlane Processing → Data Acquisition → Diagnostics & Control → Observable Measurement

Quantum Measurement Stack (QMS)

Layer	Function
Field Excitation & Imprint Formation	Quantum fields generate particle excitations and interference structure
DetectorPlane Imaging	Sensors convert optical/particle signals into electronic signals
Data Acquisition (DAQ)	Electronics digitize signals into measurable data
Quantum Diagnostics	Measurement fidelity and coherence preservation are evaluated
Quantum Control System (QCS)	Experimental parameters adjusted to stabilize measurement integrity

Detector Plane

The detector plane defines the effective measurement boundary and includes detector materials, analog electronics, digitization hardware, firmware, and software readout. Correlations lost at this stage cannot be recovered through postprocessing.

QMCTB01 Benchmark

Detector Regime	Observed Outcome
IntensityOnly Detection	Phase correlations discarded; no stable interference fringes
CorrelationPreserving Detection	Phase correlations retained; stable highcontrast fringes

Key Implications

- Irreversible information loss is localized at the detector plane.
- Interference visibility depends on detectorplane coherence fidelity.
- Measurement integrity can be treated as a controllable hardware parameter.

Research Context

This framework provides a measurementscience perspective applicable across quantum optics, singlephoton detectors, solidstate sensors, and hybrid quantum systems. The QMCTB01 benchmark enables reproducible evaluation of detector coherence fidelity across experimental platforms.